

3.2 Targets

ID	Target	Construction	Used in
T1	Registration plate	Flat, about 400 × 300 mm, with a ChArUco or circle-grid pattern that is visible in the left IR image. Pattern geometry is known to ≤ 0.02 mm.	4
T2	Noise plate	Uniform matte, NOISE_PLATE_SIZE_MM, flatness PLATE_FLATNESS_MM	A, B-Z, sentinels
T3a	Raised square	Square of side EDGE_SQUARE_SIZE_MM with knife edges, on hidden posts at G above a back plate. Spacers set G to GAP_SMALL_MM or GAP_LARGE_MM.	B-HV, E
T3b	Square window	Front plate with a square window of the same size and knife edges; back plate at G behind it	B-HV, E
T4-S / T4-L	Disk arrays, small and large	Back-beveled disks on thin posts at G above a back plate. The diameter ladder is split across two plates by size. Includes blank sites and post-only control sites.	C, D, E
T5-S / T5-L	Cutout arrays, small and large	Back-beveled holes in a front plate, with the back plate at G. Includes blank sites.	C, D, E

Diameter ladder. Diameters run from $\text{DIAMETER_MIN_FOOTPRINT_FRACTION} \times p(Z_MIN)$ up to $\text{DIAMETER_MAX_FOOTPRINT_MULTIPLE} \times p(Z_MAX)$, in steps of $\text{DIAMETER_LADDER_RATIO}$. For example, at $f_x \approx 500$ px (unconfirmed) this gives about 0.3 mm to 60 mm in about 16 steps. Features on a plate are spaced at least $\text{FEATURE_ISOLATION_PX}$ apart at Z_MIN , so the sensor's spatial interpolation cannot couple neighboring features.

Blank and control sites. Each array has blank sites matching each search-window size. These give the false-alarm rate in D. Disk arrays also have post-only sites (a post with no disk). These show whether the support post itself is detected.

Surface finish. All front and back surfaces use one finish, for example bead-blasted aluminum or a matte coating, with mid-range IR reflectance. Record the finish and, if possible, its reflectance at the projector wavelength. A reflectance difference between plates would bias the edge and area results.